

Yunwang (Luke) CHEN

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🎓 Education

Questrom School of Business, Boston University, Boston, MA, USA

Sep. 2026 – Present

Ph.D. Student in Operations and Technology Management

- › Interests: AI token pricing; energy markets; electric vehicles and sustainability; operations and market design; blockchain and automated market makers.

College of Engineering, Boston University, Boston, MA, USA

Sep. 2025 – Aug. 2026

Ph.D. Rotation Year in Systems Engineering

Fellowship: *Distinguished Systems Engineering Fellowship*

Southern University of Science and Technology (SUSTech), Shenzhen, China

Aug. 2021 - Jul. 2025

B.Eng., Department of Electrical and Electronic Engineering

GPA: 3.86/4.00

Academic Supervisor: [Prof. Perry Shum](#); Research Supervisor: [Prof. Linni Jian](#)

- › Admission through the College Entrance Examination (Gaokao) with a score of 673 points, ranked approximately 500th (top 0.07%) in Guangdong Province and admitted as the **first candidate** by SUSTech.

University of Toronto, St. George Campus (UoT), Toronto, Canada

Exchange, Aug. - Dec. 2023

Exchange Student, Faculty of Applied Science & Engineering

GPA: 4.0/4.0

- › Courses: Applied Fundamentals of Deep Learning (A), Algorithms and Data Structures (A), Matrix Algebra and Optimization (A), Mechanical Engineering Design (A).

Shenzhen Yucai Senior High School, Shenzhen, China

Sep. 2018 - Jul. 2021

- › Honor: Shenzhen Model Student

2020

📖 Commentary

Linni Jian*, **Yunwang Chen*** & Ching-chuen Chan (2026). Affordable mobility for all: why we need smaller, cheaper electric vehicles. *Nature*, 651, 585–587. *Equal contribution. [\[Commentary\]](#)

📖 Publications & Preprints

[1] **Yunwang Chen**, Yaoming Zhao & **Siuming Yiu** (2024). Cyber-Physical Authentication Scheme for Secure V2G Transactions Using Blockchain and Smart Contracts. *In Progress*. [\[Preprint\]](#)

[2] **Yunwang Chen**, Xiang Lei, Songyan Niu & **Linni Jian** (2024). Trustworthy V2G Scheduling and Energy Trading: A Blockchain-Based Framework. Published in *eTransportation*. IF: 15.0, JCR Q1. [\[Elsevier Link\]](#)

[3] **Yunwang Chen**, Yaozhong Kang, Ziqi Zhao, Yue Hong, Lingxiao Meng & **Max Q.-H. Meng** (2024). Collaborative Fall Detection and Response Using Wi-Fi Sensing and Mobile Companion Robot. Published in *IEEE International Conference on Robotics and Biomimetics (ROBIO), 2024* (Peer-reviewed). [\[Demo Video\]](#) & [\[IEEE Link\]](#)

[4] **Yunwang Chen**, Xiang Lei & **Linni Jian** (2024). Addressing Trust Issues for Vehicle-to-Grid in Distributed Power Grids Using Blockchains. Published in *International Conference on Power and Energy Systems, 2024* (Peer-reviewed). [\[IEEE Link\]](#)

[5] **Yunwang Chen**, Fan Liu, Zihan Liao & Fuwang Dong (2023). Symbol-Level Precoding for MIMO ISAC Transmission Based on Interference Exploitation. Published in *IEEE Communications Letters*. Ranked among popular works (25th & 24th) in February & March 2024. IF: 4.1, JCR Q2. [\[IEEE Link\]](#)

🏆 Honors & Awards

- › **Summa Cum Laude** — Southern University of Science and Technology 2025
- › **Joyce M. Kuok Foundation Scholarship** 2025
- › **IEEE Signal Processing Society Scholarship** [\[IEEE Newsletter\]](#) 2024
- › **SUSTech Presidential Scholarship** (Top admission candidate & sustained excellence) 2021, 2022, 2023, 2024
- › **SUSTech Merit Student Scholarship** (First Class, top 5%) 2022, 2023, 2024
- › **SUSTech 500m Rowing Ergometer Competition** (First Place, 1/203 and 1/369) 2023, 2024
- › **SUSTech Soft Fitness Challenge** (First Place, 1/564) 2023
- › **SUSTech Outstanding Volunteer** (for exemplary volunteer service during COVID19) 2022
- › **Shenzhen Model Student** 2018

⚙️ Research Experience

Ongoing Projects

- › **Centralized & Decentralized Vehicle to Grid (V2G) Business Models (Research Rotation)** Sep. 2025 – Present
Advisor: [Prof. Gerry Tsoukalas](#), Boston University
- › **V2G Market Design (Research Rotation)** Sep. 2025 – Present
Advisor: [Prof. Pengyu Qian](#), Boston University
- › **Sustainability Research on Electric Vehicles (Paper Under Review)** Nov. 2024 – Present
Advisors: [Prof. C.C. Chan](#) & [Prof. Linni Jian](#), The University of Hong Kong

Leveraging Blockchain for Trustworthy Vehicle-to-Grid Energy Trading

Jun. 2023 - Nov. 2024

Advisor: [Prof. Linni Jian](#), SUSTech EEE

- › Developed a blockchain-enabled cyber-physical framework tailored to V2G operations in the distributed power grid.
- › Formulated a game theoretical based pricing strategy to maximize the economic returns for electric vehicle aggregator and individual EV owners.
- › Demonstrated significant reductions in EV charging costs and the potential to support auxiliary grid services.

Cybersecurity in Electric Vehicle Charging and Vehicle-to-Grid Operations

Jul. 2024 - Sep. 2024

Advisor: [Prof. Siuming Yiu](#), The University of Hong Kong

- › Developed a cyber-physical authentication scheme and smart contracts to secure energy trading EV charging systems.
- › Performed security analysis in the context of V2G to mitigate Man-in-the-Middle, replay and DDoS attacks.

Collaborative Fall Detection and Response using Wi-Fi Sensing and Mobile Companion Robot

Apr. 2024 - Oct. 2024

Advisor: [Prof. Max Q.-H. Meng](#), SUSTech EEE

- › Developed a collaborative system combining Wi-Fi sensing and autonomous mobile robots for real-time assistance.
- › Built a Wi-Fi dataset of human activities in non-line-of-sight (NLOS) settings to improve detection accuracy.
- › Applied a transformer model for fall detection, achieving an 87% success rate in real-world tests.

Interference Exploitation in Integrated Sensing and Communications Systems

Sep. 2022 - Dec. 2023

Advisor: [Prof. Fan Liu](#), SUSTech EEE

- › Exploited constructive interference in downlink ISAC systems using symbol-level precoding methods.
- › Formulated optimization problems, employing semidefinite relaxation (SDR) and successive convex approximation (SCA) techniques to resolve the non-convexity.
- › Demonstrated that the proposed scheme outperformed conventional block-level precoding (BLP) systems.

Projects

Public Transport Database System

Feb. 2024 - Jun. 2024

Course Project of Principles of Database Systems, SUSTech

- › Developed a MySQL database for managing public transportation, integrating bus and subway data, with pathfinding and user access controls.
- › Built a Gradio-based web interface for real-time data interaction and management.
- › Optimized data import across systems (Windows, Mac, Linux) using Python, SQLAlchemy, and shell scripting.

Music Genre Categorization with Deep Learning

Sep. 2023 - Dec. 2023

Course Project of Applied Fundamentals of Deep Learning, UofT

 [Presentation Link](#)

- › Developed a CNN-based model using spectrograms and transfer learning with AlexNet on the GTZAN dataset for music genre classification.
- › Achieved 68.0% test accuracy and 81.4% on new data using evaluation with copyright-free tracks.

Services & Activities

Monitor of EEE Elite Student Class

Nov. 2022 - Aug. 2025

- › Led the elite student class and prepared for the seminars for elite students in the EEE Department.
- › Organized monthly class events, such as a post-finals dinner.

Reviewer

April. 2024 - Aug. 2025

- › Journal: *IEEE Communications Letters*; Conference: *IEEE Robio 2024*

SUSTech Volunteer

Sep. 2021 - Aug. 2025

- › Volunteered at various events, including international seminars and university competitions, totaling 144+ hours.

Career Planning for Freshman Students

Mar. 2022 - Sep. 2022

- › Established [SUSTecher](#), a career reading list website, to assist incoming students with university life, academics, and personal growth.
- › Worked with SUSTech library to create diverse reading promotions, fostering a culture of reading among students.
- › Completed a [digital survey report](#) on major selection at SUSTech (sophomore or junior year), featuring interactive charts to provide freshmen with valuable insights into upper-year students' decisions, experiences, and reflections.

Member of SUSTech Rowing Team

Feb. 2022 - Dec. 2022

- › Played a crucial role as the key power output in the 6th seat for the eight-man rowing team.

Skills

Languages: Mandarin (Native), English (Proficient, TOEFL iBT: 101).

Computer Skills:

- › Programming: MATLAB, Python, Java, PostgreSQL.
- › Tools and Frameworks: CVX Toolbox, PyTorch, Solidworks.

Other Skills: Rowing; Tai chi; Tennis; Swimming; Jogging; [Content Creating \(YouTuber\)](#); [Photography](#);

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